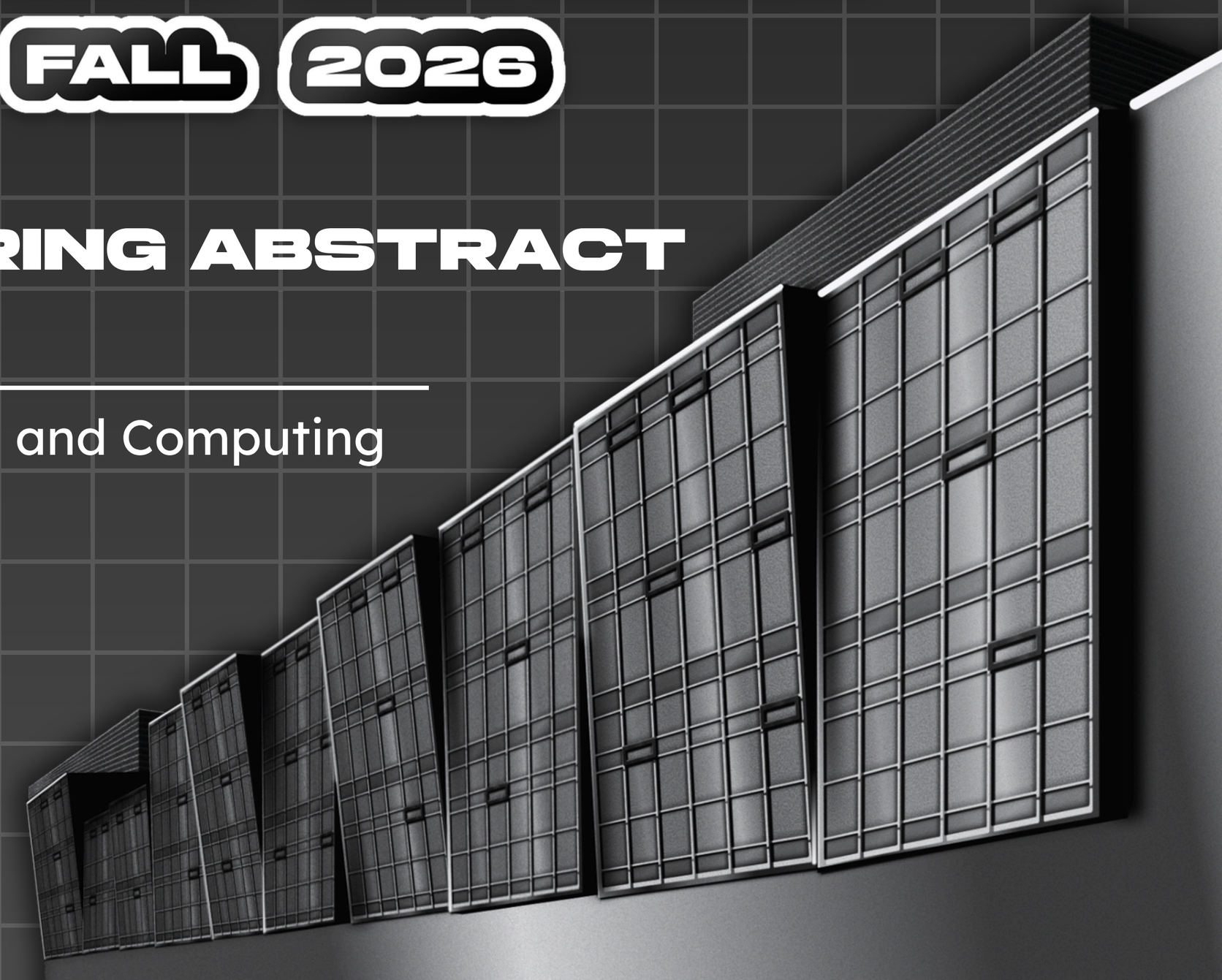
Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026



BIO-ENGINEERING ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre

1. INTRODUCTION

Welcome to the 21st Metropolitan Engineering Competition (MEC)! MEC 2026 provides students with the opportunity to apply their technical knowledge, creativity, and problem-solving skills to real-world engineering challenges while collaborating with peers and engaging with industry professionals.

The theme of MEC 2026 is Feeding the Future: Designing Smart Infrastructure for Urban Food Security. As cities continue to grow, urban food systems face increasing challenges related to limited space, environmental contamination, resource use, and long-term sustainability. Engineering and biological systems can work together to create smarter and more resilient approaches to food production.

This year's Bio-Engineering Competition, Rooting for the Future, presents competitors with a research-informed bio-engineering design challenge focused on the effects of heavy metal contamination addressing soil health and urban long-term crop productivity. Teams will apply biological research and engineering principles to inform their design and communicate their proposed solution through non-physical engineering design representations. Designs should demonstrate how biological understanding can be translated into an effective engineering system through considerations such as monitoring, implementation, maintenance, adaptability, and scalability.

2. PROBLEM STATEMENT

Heavy metal contamination from cadmium and mercury has degraded the soil in Toronto's urban tomato greenhouse, disrupting the symbiotic relationship between tomato plants and arbuscular mycorrhizal fungi (AMF), which supports water and essential nutrients uptake such as nitrogen and phosphorus.

Heavy metal exposure has inhibited fungal spore germination, reduced mycelial growth and fungal diversity, and decreased root colonization, weakening this symbiotic relationship. Declining microbial populations disrupt nitrogen and phosphorus cycling, reducing the availability of essential nutrients for plant growth. Consequently, tomato plants experience declining crop yields and increased dependence on synthetic fertilizers, contributing to water pollution, soil degradation, and threats to long-term sustainability.

Develop a practical, scalable, and cost-effective smart infrastructure solution that restores soil health, mitigates the impact of heavy metal contamination, and improves long-term crop productivity. Teams must select and justify an appropriate biological approach and develop an engineering system that supports, monitors, controls, or implements their proposed solution. Teams are not restricted to a specific biological treatment or engineering technology.

3. BACKGROUND INFORMATION

The following scenario is hypothetical and has been developed specifically for the MEC 2026 Bio-Engineering Competition. The greenhouse, contamination conditions, and associated operational data are provided for the purposes of this engineering design challenge.

As part of its 2045 Urban Food Security Initiative, the City of Toronto has invested in expanding local food production through rooftop greenhouses and urban agriculture. One of its largest projects is a 5,000 m² community greenhouse capable of producing approximately 120 tonnes of tomatoes annually for nearby grocery stores, schools, and food banks. However, production has steadily declined over the past four years despite improvements in greenhouse technology.

An environmental investigation linked the decline to contaminated runoff from a nearby former industrial site and adjacent roadway. Heavy rainfall and seasonal snowmelt introduced cadmium (Cd) and mercury (Hg) into the greenhouse soil. Soil testing identified concentrations of 3.8 mg/kg of cadmium and 1.4 mg/kg of mercury. Although the greenhouse remains operational, contamination has progressively affected soil health and conditions necessary for healthy tomato growth.

The greenhouse depends on a functioning soil ecosystem to support root development, nutrient availability, water uptake, and crop productivity. Soil microorganisms, including bacteria and fungi, play essential roles in maintaining these processes. Continued exposure to cadmium and mercury has disrupted soil function, creating increasingly difficult conditions for sustainable tomato production.

Over the same four-year period, annual tomato production has declined by approximately 24%, from 120 tonnes to 91 tonnes, reducing the supply available to nearby grocery stores, schools, and food banks by approximately 29 tonnes annually. The greenhouse has also become increasingly dependent on synthetic fertilizers to maintain production. Without intervention, continued soil degradation may further reduce crop productivity and threaten the greenhouse's long-term sustainability.

4. CONSTRAINTS

The proposed solution must meet the following constraints:

- **Pilot Area:** The initial implementation must be designed for a 250 m² section of the greenhouse.
- **Budget:** The total proposed cost of the pilot implementation must not exceed $15,000 CAD.
- **Implementation Period:** The proposed pilot must be capable of implementation within 6–12 months.
- **Greenhouse Operation:** The greenhouse must remain operational during implementation, with minimal disruption to existing tomato production.
- **Scalability:** The proposed solution must demonstrate how it could be expanded from the 250 m² pilot area to the full 5,000 m² greenhouse.
- **Monitoring:** The proposed system must include appropriate methods for monitoring relevant biological, environmental, and engineering performance indicators.
- **Maintenance and Reliability:** The design must account for realistic maintenance requirements and potential equipment or system failures.
- **Resource Efficiency:** The proposed solution should minimize unnecessary use of water, energy, fertilizers, materials, and labour.
- **Operating Costs:** Ongoing operating and maintenance costs beyond the initial pilot implementation must

be considered.

- **Environmental Impact:** The design should minimize waste, secondary pollution, and other adverse environmental impacts.
- **Safety and Compliance:** The proposed solution must account for relevant food safety, environmental, electrical, and greenhouse operating requirements. Teams must distinguish between restoring soil biological function, reducing contamination risks, improving crop productivity, and verifying the safety of harvested tomatoes.

The proposed solution must include an appropriate method for monitoring cadmium and mercury contamination and assessing whether harvested produce satisfies applicable food-safety requirements.

Teams should identify relevant regulatory guidance or standards, explain their proposed sampling and laboratory-testing approach, and describe how produce would be managed if contamination remains above acceptable limits.

Improved plant growth or increased tomato yield alone will not be considered sufficient evidence of food safety.

- **Food Safety Verification:** The proposed solution must include a method for assessing cadmium and mercury concentrations in harvested tomatoes and determining whether the produce meets applicable food safety requirements. Improvements in soil health alone must not be treated as proof that the harvested produce is safe for consumption.
- **Adaptability:** The proposed system should be capable of responding to changing soil and greenhouse conditions.
- **Testing and Validation:** The design must include measurable performance indicators and a method for evaluating the effectiveness of the pilot.
- **Long-Term Operation:** The proposed solution should consider how its performance and effectiveness can be maintained beyond the initial pilot period.

5. EXPECTED RESULTS

A successful proposed solution is expected to:

- Reduce the effects of cadmium and mercury contamination without introducing additional risks while improving soil biological function, nutrient availability, and conditions that support healthy tomato growth.
- Support an increase in annual tomato production from the current 91 tonnes toward a target of at least 110 tonnes when the proposed solution is scaled to the full 5,000 m² greenhouse. Teams are not expected to demonstrate this production increase during the competition but must justify how their proposed design could reasonably achieve it through appropriate engineering analysis, assumptions, and a testing strategy.
    - **Production Target Assessment:** Teams must explain how the expected performance of their 250 m²

pilot could support the greenhouse-wide production target when scaled to the full 5,000 m² facility. Proposed improvements must be supported by reasonable assumptions, measurable performance indicators, and a practical validation strategy.

- Demonstrate effective integration and clear justification between the selected biological approach and the proposed engineering system based on research, effectiveness, suitability, practicality, and feasibility.
- Provide measurable indicators that can be used to evaluate the performance and effectiveness of the proposed system.
- Consider relevant alternative approaches and explain the reasoning behind the final design selection.
- Provide a non-physical engineering prototype/design representation demonstrating the proposed system.

6. FINAL DELIVERABLES

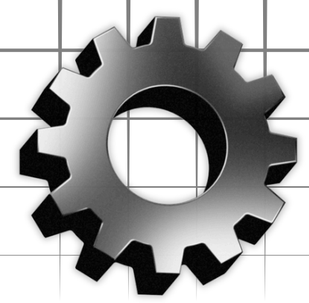
Competitors will have eight hours to analyze the problem, develop their proposed solution, and prepare their final submission. Each team must submit and present the following deliverables:

1. Final Engineering Report

Teams must submit a final engineering report with a maximum length of six pages, excluding the title page and references. The report must be submitted as a PDF. The title page must include the team name, competitor names, competition division, and proposed solution title. All external sources, figures, data, and referenced technical information must be appropriately cited using a consistent citation format.

The report must include:

- Analysis of the biological and engineering problem.
- Explanation and justification of the selected biological approach.
- Description of the proposed engineering system and its integration with the biological approach.
- Relevant engineering calculations, assumptions, and technical analysis.
- Comparison of alternative approaches and design trade-offs.
- Non-physical engineering prototype or design representations.
- Proposed testing, validation, and monitoring methods.
- Implementation plan, including the 250 m² pilot and potential expansion to the full 5,000 m² greenhouse.
- Estimated pilot implementation costs within the $15,000 CAD budget and consideration of ongoing operating costs.
- Maintenance, reliability, safety, environmental, and sustainability considerations.
- Limitations, risks, and supporting references.



2. Non-Physical Engineering Prototype or Design Representation

Teams must develop at least one detailed non-physical engineering representation that clearly demonstrates how their proposed system would function.

The representation must demonstrate the proposed system's major components, their functions, and how they interact to support the selected biological treatment. Teams may use annotated diagrams, technical specifications, simulations, control logic, or other appropriate evidence to explain how their design would operate. A fully functional software application or simulation is not required unless it is central to the team's proposed solution.

Acceptable representations include, but are not limited to:

- CAD models and engineering drawings.
- System architecture or block diagrams.
- Circuit schematics and PCB designs.
- Simulations and digital models.
- Process flow diagrams.
- Software applications, websites, or user-interface prototypes.
- Control logic and monitoring dashboards.

Physical prototypes are not required or permitted as part of the competition submission.

3. Final Presentation

Each team must prepare and submit presentation slides in PDF format and deliver a 10-minute presentation explaining its proposed solution, biological rationale, engineering design, implementation strategy, and expected outcomes. The presentation should include appropriate technical visuals, such as system diagrams, CAD models, simulations, or other non-physical engineering representations. Presentation slides must be submitted by the official competition deadline communicated by the Competition Coordinators.

4. Judge Question-and-Answer Session

Following the presentation, teams will participate in a five-minute question-and-answer session with the judges.

Teams should be prepared to explain their engineering decisions, biological concepts, design trade-offs, technical feasibility, and methods for evaluating the effectiveness of their proposed solution.